Exploring URL Shortner System Design

Day 110
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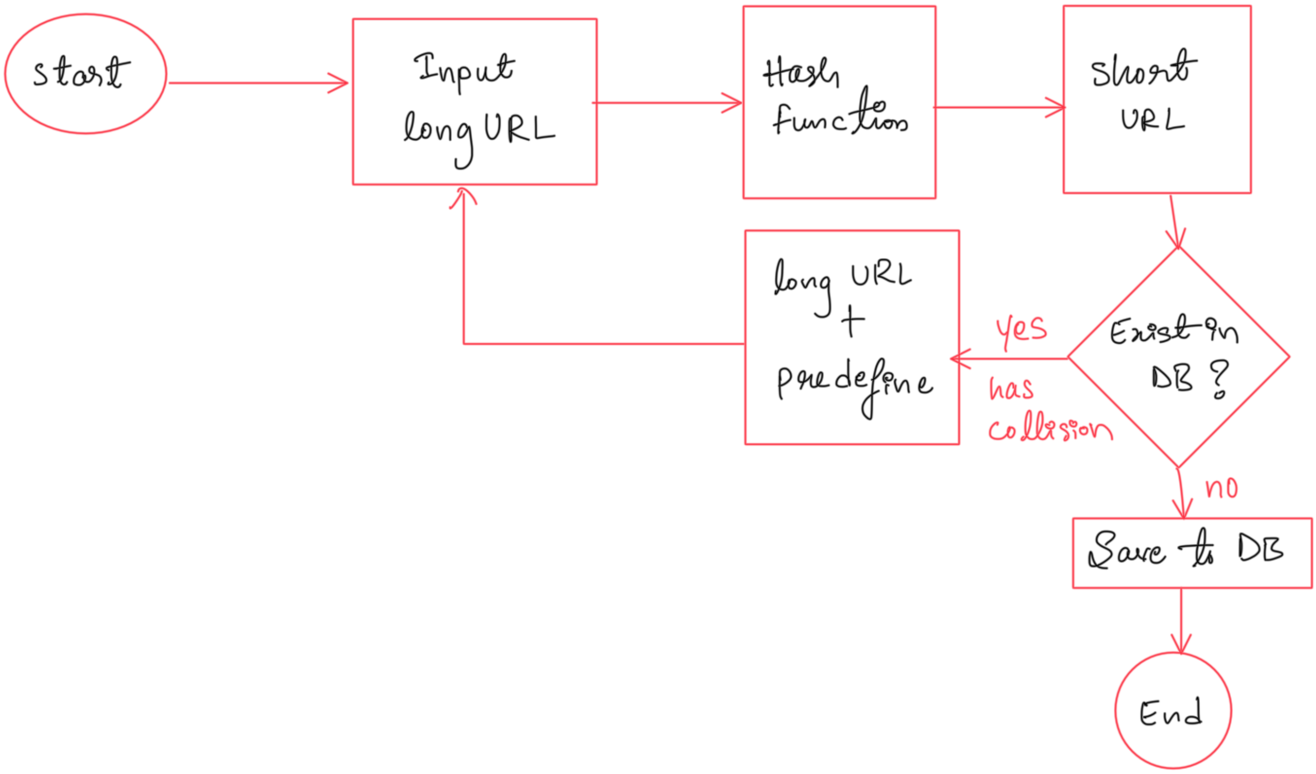
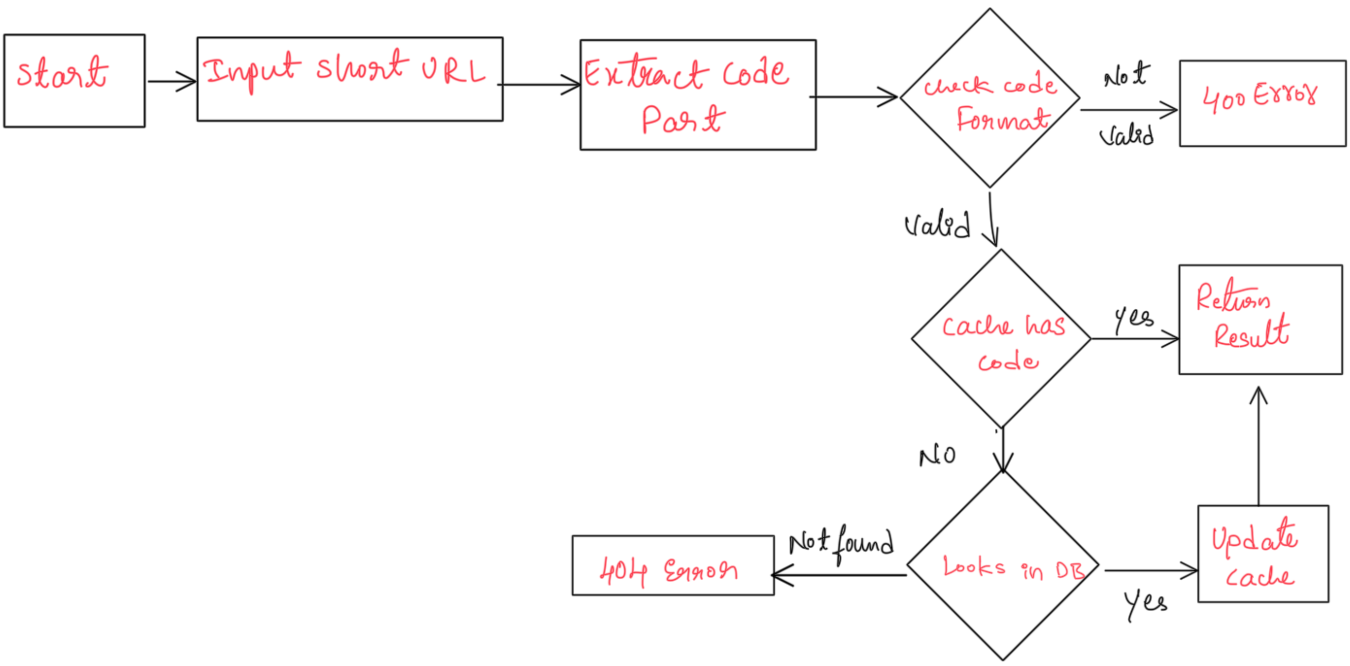
The Need for an Effient and concise URL Management system has become significant concern in the digital age.

URL shotening Services, such as bit.ly, TinyURL, and ZipZy.in, play a Massive role in transforming lengthy web addresses into shortes sharable links.





Requirements for URL shortner Service System Design

1. Functional Requirements

- Given a long URL, the Service should generate a shorter and Unique alias for it
- When the user hits a short link, the Service should redirect to the original link
- Links will Expire after a standard default time span

2. Non - Functional requirements

- The System should be highly available. This is really important to consider because if the Service goes down, all the URL redirection will start failing.

- URL redirection should happen in real-time with minimal latency
- shortened links should not be predictable

Capacity Estimation

Lets assume our Service has 30M new URL shortenings per Month. Lets assume we store Every URL shortening request for 5 years. For this period the Service will generate about 1.8 B records.

$$\boxed{30 \text{ million} * 5 \text{ years} * 12 \text{ Months} = 1.8 B}$$

NOTE :- Let's Consider we are using 7 characters to generate a short URL These characters are a combination of 62 characters [A-Z, a-z, 0-9] Something like https://zpzy.in//redirecting?code = abxdef2.

Data Capacity Modeling

Discuss the data capacity model to estimate the storage of the system. We need to understand how much data we might have to insert into our system. Think about the different columns or attributes that will be stored in our database and calculate the storage of data for five years. Let's make the assumption given below for different attributes.

- Consider the average long URL size of 2KB ie for 2048 characters.
- Short URL size: 17 Bytes for 17 characters
- created_at- 7 bytes
- expiration_length_in_minutes -7 bytes

The above calculation will give a total of 2.031KB per shortened URL entry in the database.
If we calculate the total storage then for 30 M active users
*total size = 30000000 * 2.031 = 60780000 KB = 60.78 GB per month. In a Year of 0.7284 TB and in **5 years 3.642 TB** of data.*

Note: We need to think about the reads and writes that will happen on our system for this amount of data. This will decide what kind of database (RDBMS or NoSQL) we need to use.

Tomorrow I will Explore about URL shortner System design